MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics
Class: Class X (CBSE)

WEEKLY CHAPTER TEST – 04
Chapter 13: Surface Areas and Volumes

Time: 45 Minutes
MM: 20

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 objective questions (1 mark each).
2. Section B has 2 Short Answer questions (2 marks each). Section C has 2 questions (3 marks each).
3. Section D has 1 Long Answer HOTS question (6 marks). Show all steps clearly.
4. Take $\pi = \frac{22}{7}$ unless specified otherwise. Show all steps and calculations clearly.

SECTION A: MULTIPLE CHOICE QUESTIONS (1 Mark Each) [4 Marks]

1. If two solid hemispheres of the same base radius r are joined together along their bases, then the curved surface area of this new solid is: **[1]**

(a) (a) $4\pi r^2$ (b) $6\pi r^2$ (c) $3\pi r^2$ (d) $8\pi r^2$

2. A spherical ball of radius r is melted to make 8 new identical balls each of radius r_1. Then r_1 is equal to: **[1]**

(a) (a) r (b) $\frac{r}{2}$ (c) $\frac{r}{4}$ (d) $2r$

3. Water flows at the rate of 10 metres per minute through a cylindrical pipe of internal diameter 5 mm. How long will it take to fill a conical vessel of base diameter 40 cm and depth 24 cm? **[1]**

(a) (a) 48 min 15 s (b) 51 min 12 s (c) 52 min 10 s (d) 53 min

4. **Assertion (A):** If the surface areas of two spheres are in the ratio 16 : 9, then their volumes are in the ratio 64 : 27.
Reason (R): If S_1, S_2 are surface areas of two spheres and V_1, V_2 are their volumes, then $\frac{V_1}{V_2} = \left(\frac{S_1}{S_2}\right)^{3/2}$. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

5. Rasheed got a playing top (lattu) as his birthday present, which had no colour on it. The top is shaped like a cone surmounted by a hemisphere. The entire top is 5 cm in height and diameter is 3.5 cm. Find the area he has to colour. **[2]**

6. A decorative block is made of two solids — a cube and a hemisphere. The base is a cube with edge 5 cm, and the hemisphere fixed on top has diameter 4.2 cm. Find the total surface area of the block. **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

7. The difference between the outside and inside curved surface areas of a cylindrical metallic pipe 14 cm long is 44 cm^2. If the pipe is made of 99 cm^3 of metal, find the outer and inner radii of the pipe. **[3]**

8. A wall of length 24 m, thickness 0.4 m, and height 6 m is constructed with bricks, each of dimensions 25 cm $\times$ 16 cm $\times$ 10 cm. If mortar occupies $\frac{1}{10}$th of the total volume of the wall, find the number of bricks used. **[3]**

SECTION D: LONG ANSWER / HOTS QUESTION (6 Marks) [6 Marks]

9. Spherical marbles of diameter 1.4 cm are dropped into a cylindrical beaker of diameter 7 cm containing some water.

(i) Find the volume of water displaced when the water level rises by 5.6 cm.
(ii) Calculate the volume of one spherical marble.
(iii) Determine the number of marbles that must be dropped to achieve this rise in water level.

[6]

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— **END OF QUESTION PAPER** —

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